

Magnon–crystal-field cross peaks in two-dimensional terahertz spectroscopy of TmFeO₃: two symmetry-selected vertices and a polarization-dark crystal-field dipole

Working draft¹

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We construct a complete microscopic theory of two-dimensional coherent terahertz spectroscopy (2DCS) in the rare-earth orthoferrite TmFeO₃, treating the Fe³⁺ spins classically and the three lowest Tm³⁺ crystal-field (CEF) singlets as an SU(3) qutrit whose Bloch dynamics is quantum-exact at single-ion level. Projecting all symmetry-allowed Fe–Tm couplings through the *Pbnm* space group and time reversal leaves exactly two active conversion vertices: a static two-magnon exchange $W S^2 \lambda$ and a field-gated $\kappa^B B S \lambda$. Driven by the measured single-cycle THz waveform, with every damping rate set by a measured linewidth and every emission weight by a measured oscillator strength, one Hamiltonian reproduces the peak inventories of the three detected channels without retuning: the magnetoelectric (q_{AFM}, E_{12}) peak for $H \parallel a$, the magnon–magnon ($q_{\text{FM}}, q_{\text{AFM}}$) and gated (E_{12}, q_{AFM}) peaks for $H \parallel c$, and the same-polarization CEF family (E_{12}, E_{23}), ($E_{12}, 0$), (E_{12}, E_{13}). The same-polarization hierarchy—both transfer peaks non-rephasing, with (E_{12}, E_{23}) dominant—follows from a single selection rule: the $1 \rightarrow 3$ CEF transition is electric-dipole dark for $E \parallel c$, forcing the transfer through a coherence $\rightarrow$ population $\rightarrow$ coherence pathway. We give falsifiable predictions for each mechanism.

Introduction.—Two-dimensional coherent THz spectroscopy resolves the nonlinear couplings between collective modes that one-dimensional absorption cannot see: a cross peak at (ω_τ, ω_t) certifies that a coherence prepared at ω_τ was converted into an emitter at ω_t [1, 2]. In rare-earth orthoferrites the available modes span two very different subsystems—the Fe magnons and the rare-earth crystal-field (CEF) transitions—so each cross peak is a direct measurement of a specific Fe–rare-earth coupling term. TmFeO₃ is an ideal testbed: below the spin-reorientation transition the Fe sublattice orders in Γ_2 (F_x, C_y, G_z) [3–5], the quasi-ferromagnetic and quasi-antiferromagnetic magnons sit at $q_{\text{FM}} = 0.38$ and $q_{\text{AFM}} = 0.90$ THz, and the three lowest Tm CEF singlets form a qutrit with transitions $E_{12} = 0.50$, $E_{23} = 0.70$, $E_{13} = 1.20$ THz, all within one single-cycle THz pulse bandwidth. Experiments in three detection channels—cross-polarized for $H \parallel a$ and $H \parallel c$, and same-polarized for $H \parallel a$ —observe distinct, reproducible cross-peak inventories. The theoretical question is sharp: which microscopic couplings, allowed by the *Pbnm* space group and time reversal, produce exactly these peaks and no others?

Here we answer it constructively. Three physical rules, each traceable to symmetry or to a measured matrix element, organize every observed feature: (i) of all symmetry-allowed Fe–Tm couplings, exactly two can convert coherences between the magnon and CEF sectors; (ii) the CEF transitions are electric-dipole (E1) bright while the q_{AFM} magnon is E1-dark but magnetic-dipole (M1) bright, so the readout mechanism—not the Hamiltonian—differs between geometries; (iii) in the $E \parallel c$ polarization the $1 \rightarrow 3$ CEF dipole vanishes, a parity selection rule that fixes the entire same-polarization hierarchy including the signed (rephasing versus non-

rephasing) sides.

Model.—The Fe³⁺ ($S = 5/2$) sublattice is described by classical spins with the calibrated orthoferrite Hamiltonian (exchange, single-ion anisotropy, Dzyaloshinskii–Moriya), reproducing q_{FM} , q_{AFM} , and the Γ_2 ground state. Each Tm³⁺ (3H_6 , 13 CEF singlets in C_s site symmetry) is truncated to its three lowest singlets and represented by the Gell-Mann coordinates $\lambda^a = \langle \Lambda^a \rangle$; because the projected single-ion Hamiltonian is linear in the Λ^a , the classical $\vec{\lambda}$ equation of motion is the exact von Neumann equation of the driven, damped three-level system, and a Boltzmann ensemble over level seeds is the exact thermal density matrix [SI, Sec. S1]. The Fe–Tm sector is constructed by projecting the physical exchange $H = \mathbf{S}^T \mathbf{A} \mathbf{J}$ into the qutrit (PJP) in a transported gauge that makes all four Tm sublattices explicit [SI, Secs. S2–S3].

Every candidate coupling is then scored against six requirements (reach the bright coordinate; be sourced by the pumped magnon; respect time-reversal parity; be Γ_1 -allowed in Γ_2 ; carry the frequency mismatch; involve both pulses). The result [SI, Sec. S4] is exhaustive: the static bilinear $K^- S \lambda^-$ fails threefold—as a linear hop it hybridizes but cannot convert, its channels are Γ_1 -cancelled to machine precision, and its one allowed component reorients the ground state. The complete quadratic families W_4 , W_1^{xy} , W_6 , κ^E are exactly inert. Two vertices survive,

$$H_W = W_1^{yy} S_y^2 \lambda^1, \quad H_\kappa = \kappa_{1y}^B B_y(t) S_y \lambda^1, \quad (1)$$

and both are required by the data, as shown below.

Observable.—Two collinear pulses $B_A(t + \tau)$, $B_B(t)$ give $M_{\text{NL}} = M_{AB} - M_A - M_B$, which cancels all single-pulse terms and starts at the mixed second order; the experimental scaling is $E_{\text{exc}}^2 E_{\text{det}}^1$ (third order). The

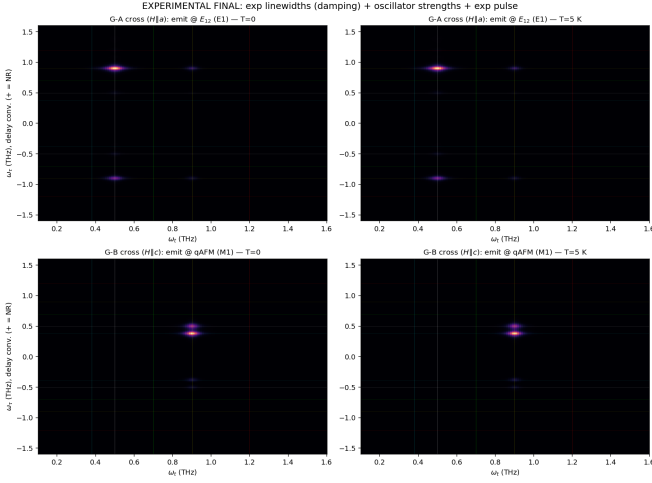


FIG. 1. Cross-polarized channels computed from experimental inputs (measured pulse, linewidth damping, oscillator-strength emission). Top: geometry A ($H \parallel a$), Tm λ^2 channel emitting at E_{12} ; the magnetoelectric (q_{AFM}, E_{12}) cross peak dominates (rephasing partner 0.33). Bottom: geometry B ($H \parallel c$), magnon M1 channel emitting at q_{AFM} ; the magnon-magnon (q_{FM}, q_{AFM}) and gated transfer (E_{12}, q_{AFM}) peaks. Left $T = 0$, right $T = 5$ K.

delay axis is plotted physically, $+\omega_\tau =$ non-rephasing (NR), $-\omega_\tau =$ rephasing (R). All spectra below are computed with the *measured* single-cycle pump (spectrum peaked at $0.53 \text{ THz} \approx E_{12}$, no DC), each coherence damped at its measured linewidth [$\gamma(E_{12}, E_{13}, E_{23}) = 0.038, 0.05, 0.10 \text{ THz}$ Bloch; magnon Gilbert $\alpha = 1.7 \times 10^{-3}$], and each emitter weighted by its measured oscillator strength ($\sqrt{f} = 2.5\text{--}3.0$ for the CEF E1 lines; $f_{q_{AFM}} = 7 \times 10^{-4}$, E1-dark) [SI, Sec. S5]. Peaks are identified blindly as 2D local maxima with prominence ≥ 1.5 over the local median background.

Geometry A, cross-polarized: a magnon read out through a CEF dipole.—During τ the Zeeman-launched q_{AFM} magnon precesses; the static vertex W_1^{yy} , by which the magnon distortion modulates the CEF splitting, transcribes the magnon phase onto the bright E_{12} coordinate. The two-pulse cross term of S_y^2 rectifies,

$$M_{NL}^W \propto W_1^{yy} A^2 \cos(q_{AFM}\tau) [1 - \cos E_{12}t], \quad (2)$$

a factorized, rank-1 cross peak at (q_{AFM}, E_{12}) [Fig. 1, top]. The E_{12} quantum is supplied by the pulse-edge bandwidth (an intra-pulse difference-frequency process): stretching the pulse from $\sigma = 0.25$ to 1.0 at fixed amplitude collapses the peak by 4.5 decades, the adiabatic suppression this displacive mechanism demands, while switching off the direct Tm drive changes it by $< 10^{-4}$ [SI, Sec. S6]. The peak is a *magnetoelectric* object: the E1-dark magnon borrows the Tm E1 dipole and emits at E_{12} , not at its own frequency.

Geometry B, cross-polarized: M1 readout and the two-vertex division of labor.—For $H \parallel c$ both observed peaks

emit at q_{AFM} through the magnon M1 dipole. The magnon-magnon peak (q_{FM}, q_{AFM}) requires no Fe-Tm physics (anisotropy plus DM couple the branches). The transfer peak (E_{12}, q_{AFM}) isolates the second vertex: with κ_{1y}^B alone it is present (0.46); with the static hybridization alone it is absent. The field gate acts only while $B_z(t) \neq 0$, imprinting the stored E_{12} phase impulsively on the magnon—and the same gate makes κ^B *silent* in geometry A ($B_z = 0$), where its background would otherwise contaminate the clean W peak. Conversely, the measured post-drive buildup of the geometry-B signal is impossible for a gated vertex and requires the static W_1^{xz} hybridization, whose slow exchange keeps pumping the magnon after the field is gone [SI, Sec. S7]. Neither vertex alone survives the data; together they reproduce $(q_{FM}, q_{AFM}) = 1.00$, $(E_{12}, q_{AFM}) = 0.43$, residuals ≤ 0.12 [Fig. 1, bottom]. The Tm CEF channels are identically zero here: the $E \parallel a$, $H \parallel c$ drive closes on the $\{\lambda^{1,2,3}\}$ subalgebra.

Geometry A, same-polarized: a dark dipole fixes the hierarchy.—The same-polarization channel ($E \parallel c$ in and out) shows five features, in order $(E_{12}, E_{23}) \approx (E_{12}, 0) > (E_{12}, E_{13}) > (q_{AFM}, 0.38), (q_{AFM}, E_{13})$, with *both* CEF transfer peaks on the non-rephasing side. This inventory contains a structural clue: there is no peak with delay phase $E_{13} = 1.2 \text{ THz}$ anywhere in the map. A bright $1 \rightarrow 3$ dipole would necessarily store an E_{13} delay coherence at first order and produce an (E_{13}, E_{23}) peak with *exactly* the same dipole product $\mu_{12}\mu_{13}\mu_{23}$ as the observed (E_{12}, E_{23}) peak—no parameter choice can separate them. The absence of the whole row is therefore a selection rule,

$$\mu_{13}^z \approx 0, \quad (3)$$

the natural consequence of a parity-alternating CEF ladder ($|1\rangle, |3\rangle$ even, $|2\rangle$ odd under the site mirror): the z -dipole connects $1 \leftrightarrow 2$ and $2 \leftrightarrow 3$ only. The measured E_{13} oscillator strength ($f = 8.7$) lives in the orthogonal polarizations—the same transition is bright in one polarization and dark in the other.

Equation (3) fixes three observables at once [Fig. 2]. (i) With the single-action (rephasing) conversion route $\rho_{12} \rightarrow \rho_{32}$ removed, the dominant transfer runs coherence $\rightarrow$ population $\rightarrow$ coherence: one probe action converts the stored E_{12} coherence into a level-2 population still carrying the delay phase, a second converts it into the emitting E_{23} coherence. This route is co-rotating and lands *non-rephasing*, the observed side; taken at its zero-frequency output it also produces the $(E_{12}, 0)$ peak, a sharp E_{12} line in the rectified signal $S_{DC}(\tau)$. (ii) The partner (E_{12}, E_{13}) , though lower order, emits through the throttled μ_{13}^z and is capped at 0.78 of the dominant peak (the measured ratio calibrates the residual admixture at the percent level). (iii) The magnon-delay peak (q_{AFM}, E_{13}) is weak and thermally activated ($0.10 \rightarrow 0.51$

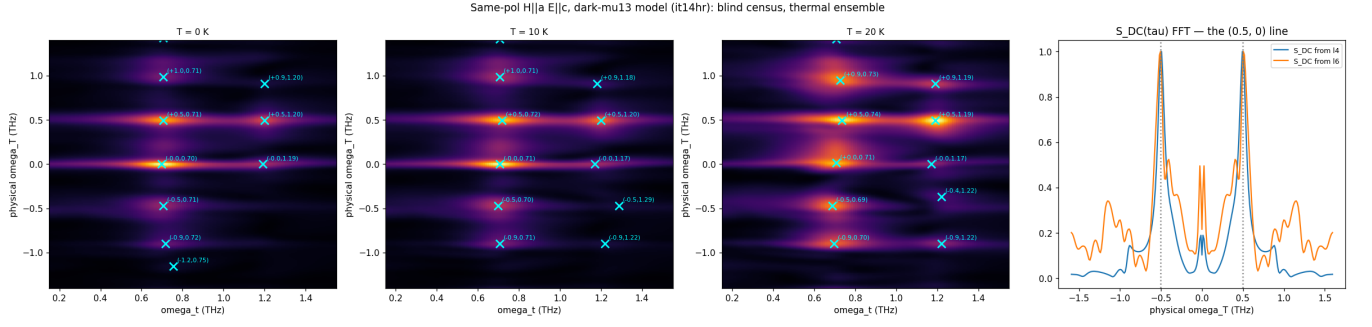


FIG. 2. Same-polarized geometry-A channel (blind census at 0, 10, 20 K; crosses mark genuine local maxima). Both transfer peaks, $(+E_{12}, E_{23})$ and $(+E_{12}, E_{13})$, are non-rephasing, with the E_{23} member dominant; the rephasing remnant at $(-E_{12}, E_{23})$ and the two-quantum line at $\omega_\tau = 2E_{12}$ are predictions. Right: the rectified signal $S_{DC}(\tau)$ whose sharp line at $+E_{12}$ is the $(E_{12}, 0)$ peak.

from 0 to 20 K). The one feature the model does not produce is $(q_{AFM}, 0.38)$: the q_{FM} emission exists (a b -axis M1 ridge, as Γ_2 symmetry requires) but no $q_{AFM} \rightarrow q_{FM}$ transfer vertex of sufficient strength exists in the present Hamiltonian class [SI, Sec. S8].

Because the single-ion dynamics is quantum-exact, these statements were verified against a standalone qutrit integrator with the same pulse and dampings, which reproduces the lattice census exactly and shows two robust residuals that we advance as predictions: a rephasing remnant at $(-E_{12}, E_{23})$ at ~ 0.5 of the dominant peak (an exact single-ion floor under this pulse), and a lattice two-quantum line at $\omega_\tau = 2E_{12} = 0.99$ THz (absent in the single-ion dynamics, i.e. a mean-field/inter-site product) [SI, Sec. S8].

Predictions.—Each mechanism carries its own falsifiable signature. (1) *Emission fingerprint*: the geometry-A cross peak emits at a CEF frequency, the geometry-B peaks at the magnon frequency; any peak emitting at q_{AFM} inherits the $15\times$ sharper magnon linewidth along ω_t . (2) *Thermal trends*: (q_{FM}, q_{AFM}) is temperature-flat; (E_{12}, q_{AFM}) falls as the population difference $n_1 - n_2$; (q_{AFM}, E_{13}) grows steeply; the same-polarization pair approaches parity by ~ 20 K. (3) *Polarization test of the dark dipole*: rotating either polarization away from c must revive the $\omega_\tau = 1.2$ THz delay rows. (4) *Field scaling*: the dominant (E_{12}, E_{23}) peak is one field order above (E_{12}, E_{13}) ; lowering the fluence inverts their order. (5) *Sub-features*: the rephasing remnant at $(-E_{12}, E_{23})$ and the two-quantum line at $\omega_\tau = 0.99$ THz (resolvable from the 0.90 magnon row) should both be present in raw data. (6) *Pulse-length test*: the geometry-A cross peak is displacive and collapses for adiabatically long pulses.

Conclusion.—A single Fe–Tm Hamiltonian with two symmetry-selected vertices, driven by the measured pulse and dressed with measured linewidths and oscillator strengths, accounts for the complete 2DCS peak inventory of $TmFeO_3$ across three detection channels. The analysis illustrates a general program for magnon–crystal-field 2DCS: time-reversal and irrep bookkeeping reduce the coupling space to a few vertices; emission dipoles (E1 versus M1) decide where each coherence is read out; and polarization-resolved selection rules—here a z -dark $1 \rightarrow 3$ dipole—control the pathway orders and hence the signed-side structure of the 2D map. The framework, and the quantum-exactness of the $SU(3)$ single-ion dynamics, carry over directly to other rare-earth orthoferrites and to singlet-ladder magnets generally.

Numerical details, derivations, and the complete falsification ledger are given in the Supplemental Material.

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